

Marlin-Flavour G-code Syntax Reference

Purpose: practical command reference for a C#.NET Grasshopper component that converts ordered RhinoCommon curves into Marlin-compatible G-code.

Scope: commands needed to generate, validate, preview, and safely export a normal motion/extrusion job. Marlin contains many additional calibration, networking, display, laser, CNC, and board-specific commands; consult the [official Marlin G-code index](#) for the complete firmware catalogue.

Safety: Never assume that a command supported by one Marlin printer is enabled on another. Start/end code, temperature limits, homing, bed levelling, cold-extrusion behaviour, tool selection, motion limits, and parking coordinates must come from a tested `MachineProfile` for the exact printer.

1. G-code line syntax

General form:

```
G1 X120.500 Y45.250 Z8.000 E14.627 F1800 ; deposition move
```

Element	Meaning	Generator rule
G	Preparatory/motion command	Use for moves, units, coordinates, homing and dwell.
M	Machine/firmware command	Use for extrusion modes, heaters, fans, limits and status.
T	Tool/extruder selection	Emit only for a verified multi-tool machine profile.
X Y Z	Axis destination	Millimetres when <code>G21</code> is active. Absolute under <code>G90</code> ; relative under <code>G91</code> .
E	Extruder-axis destination or increment	Absolute under <code>M82</code> ; relative under <code>M83</code> .
F	Feedrate	Current units per minute; normally mm/min. It remains active until changed.
S	Common value parameter	Meaning depends on the command: temperature, fan PWM, percentage, etc.
P	Common secondary parameter	Often time, index or a command-specific value.
I J K	Arc-centre offsets / command-specific parameters	For <code>G2/G3</code> , normally arc-centre offsets from the start point.
R	Radius or wait-both-directions temperature	Meaning depends on the command. Never interpret without command context.
;	Comment start	Everything after <code>;</code> is a human-readable comment.
N...*...	Optional line number and checksum	Mainly for host streaming protocols; not normally required in an SD-card file.

Formatting rules for the C# postprocessor

- Emit one command per line using `\n` or the selected output newline policy.
- Use `CultureInfo.InvariantCulture` ; decimal separators must be `.` .
- Never emit `NaN` , infinity, empty coordinates, or locale-formatted numbers.
- Use consistent precision, commonly 3 decimals for XYZ and 4-5 for E.
- Remove unnecessary trailing zeros only if the output remains unambiguous.
- Do not repeat modal values unless readability or the machine profile requires it.
- Treat feedrate as state: `G1 F1800` affects following moves until another `F` is supplied.
- Keep start protocol, generated motion, and end protocol as separate blocks internally.

Example C# formatter:

```
using System.Globalization;

static string Mm(double value, int decimals = 3) =>
    value.ToString($"0.{new string('#', decimals)}", CultureInfo.InvariantCulture);

string line = $"G1 X{Mm(x)} Y{Mm(y)} Z{Mm(z)} E{Mm(e, 5)} F{feedMmPerMin}";
```

2. Modal machine state

The generator must explicitly establish these modes near the beginning of every file.

State	Recommended declaration	Meaning
Units	G21	Millimetres. Do not rely on firmware startup state.
XYZ positioning	G90	Absolute X/Y/Z positions from Rhino/Grasshopper coordinates.
Extrusion positioning	M83 or M82	Relative or absolute E; choose exactly one through <code>MachineProfile</code> .
Extruder origin	G92 E0	Reset logical E position, mainly useful with absolute extrusion.
Active tool	T0 if required	Select first tool on verified multi-tool machines.
Levelling	M420 S1 or machine-specific G29	Restore or generate bed compensation after homing, when supported.

Important interactions:

- G90/G91 controls XYZ and may also affect E unless M82/M83 explicitly overrides extrusion mode.
- Always emit M82 or M83 after choosing G90/G91 so extrusion behaviour is unambiguous.
- G28 may disable bed levelling. Restore it after homing if the printer requires M420 S1.
- G92 changes the logical position without moving the machine.

3. Core commands the generator must understand

3.1 Motion and coordinate commands

Command	Typical syntax	Description	Generator use
G0	G0 X100 Y50 Z5 F4200	Linear non-deposition/rapid move. Marlin generally queues it like G1; on SCARA it may have different behaviour.	Travel, approach, Z-hop and parking moves. Omit E.
G1	G1 X100 Y50 Z2 E0.842 F1800	Coordinated linear move. E synchronises extrusion with XYZ movement.	Primary deposition command. Also used for controlled retract/prime moves.
G2	G2 X50 Y20 I10 J0 E1.25 F1800	Clockwise arc/circle move when ARC_SUPPORT is enabled.	Optional arc postprocessor. Validate plane and firmware support.
G3	G3 X50 Y20 I10 J0 E1.25 F1800	Counter-clockwise arc/circle move when ARC_SUPPORT is enabled.	Optional arc postprocessor. Fall back to sampled G1 segments.
G4	G4 P500 or G4 S1	Dwell. P is milliseconds; S is seconds.	Process pause, pressure stabilisation or timed wait.
G21	G21	Select millimetres.	Required near file start.
G28	G28 / G28 X Y	Home all or selected axes. Establishes a trusted machine position.	Start protocol only, using machine-approved axes/order.
G29	Variant-specific	Probe or operate bed levelling. Parameters vary by configured levelling system.	Never invent generic parameters. Copy verified printer start code.
G90	G90	Absolute positioning.	Recommended for curve coordinates.

Command	Typical syntax	Description	Generator use
G91	G91	Relative positioning.	Useful temporarily for safe Z-lift/retraction sequences. Restore G90.
G92	G92 E0	Set the current logical coordinate without moving.	Reset E or establish a deliberate work coordinate.

3.2 Extrusion mode and flow commands

Command	Typical syntax	Description	Generator use
M82	M82	Absolute extruder positioning. Every E value is cumulative.	$E[n] = E[n-1] + \text{segmentExtrusion}$. Periodically reset using G92 E0 if required.
M83	M83	Relative extruder positioning. Every E value applies to the current move.	Simplest mode for a custom curve generator. $E = \text{segmentExtrusion}$.
M200	M200 D1.75 / M200 D0	Enable volumetric extrusion by filament diameter, or disable with D0.	Use only if the machine profile explicitly expects volumetric E.
M221	M221 S100	Global extrusion-flow percentage applied to E moves.	Optional process override. Prefer calculating correct E directly.
G10	G10	Firmware retract when FWRETRACT is enabled.	Optional alternative to explicit negative-E moves.
G11	G11	Firmware recover/unretract.	Pair with G10; requires verified firmware setup.
M207	Profile-specific	Configure firmware retract distance and speed.	Configuration/profile command, not normally repeated in every file.
M208	Profile-specific	Configure firmware recover values.	Configuration/profile command, not normally repeated in every file.
M302	M302 S0	Allows cold E movement by changing/disabling minimum extrusion temperature.	Clay/paste only when explicitly approved. Dangerous in an FDM profile.

Filament-extrusion starting formula:

```

depositedVolume = segmentLength * beadWidth * layerHeight
filamentArea    = pi * filamentDiameter^2 / 4
segmentE        = depositedVolume / filamentArea * flowMultiplier

```

For clay, paste, concrete, chocolate, syringe or pump extrusion, use a process-specific `IExtrusionModel`. Do not apply the filament-area formula unless E genuinely represents incoming filament length.

3.3 Temperature commands

Command	Typical syntax	Description	Generator use
M104	M104 S200	Set hotend target and continue without waiting.	Begin nozzle heating in parallel with other safe setup actions.
M109	M109 S200	Set hotend target and wait when heating. R waits for heating or cooling.	Required before FDM extrusion unless validated otherwise.
M105	M105	Report current and target temperatures.	Host diagnostics; generally unnecessary inside SD-card G-code.
M140	M140 S60	Set bed target and continue without waiting.	Begin bed heating early.
M190	M190 S60	Set bed target and wait when heating. R waits for heating or cooling.	FDM start protocol when a heated bed is used.
M104 S0	M104 S0	Set active hotend target to zero.	End protocol heater shutdown.

Command	Typical syntax	Description	Generator use
M140 S0	M140 S0	Set bed target to zero.	End protocol bed shutdown.

M109 and M190 can block command processing while waiting. Hosts can only interrupt reliably when the firmware is configured appropriately. Do not emit temperatures outside the MachineProfile limits.

3.4 Fan commands

Command	Typical syntax	Description	Generator use
M106	M106 S128	Set default print-fan PWM from 0 to 255. P selects a fan index.	Material/layer-dependent cooling. S255 is full duty.
M107	M107	Turn off the selected/default fan.	Start and end protocol, or process-specific cooling changes.

3.5 Motion limits and overrides

Command	Typical syntax	Description	Generator use
M201	Profile-specific	Set axis acceleration limits.	Machine configuration; avoid overriding without approval.
M203	Profile-specific	Set maximum feedrates by axis.	Machine configuration; validate against it rather than rewriting it.
M204	M204 P800 R1200 T1500	Set printing, retract and travel acceleration.	Optional job-level tuning from a verified profile.
M205	Profile-specific	Advanced motion settings: junction deviation or classic jerk, minimum rates and segment time.	Firmware/configuration-dependent. Do not emit generic values.
M220	M220 S100	Global feedrate percentage applied to G-code moves.	Optional reset to 100% at start; machine policy decides.
M221	M221 S100	Global E-flow percentage.	Optional reset to 100% at start.
M400	M400	Wait until every queued move is complete.	Before shutdown, tool change, beep, pressure stop or final state change.

3.6 Bed levelling and coordinate compensation

Command	Typical syntax	Description	Generator use
M420	M420 S1	Enable previously stored bed-levelling compensation when supported.	Usually after G28 ; exact behaviour depends on configuration.
G29	Printer-specific	Probe, generate, load or manipulate a levelling mesh depending on the configured system.	Copy known working start code. Do not guess.
M851	Profile-specific	Set probe offset.	Calibration only; never generated per job.
G53	G53 G0 X0 Y0	Perform a move in native machine coordinates when enabled.	Advanced parking only with verified firmware/profile support.

3.7 Tool, steppers, job state and messages

Command	Typical syntax	Description	Generator use
T0 - T7	T0	Select active tool/extruder.	Multi-tool profile only. Tool changes need associated temperature, offset, purge and travel logic.

Command	Typical syntax	Description	Generator use
M17	M17	Enable stepper motors.	Usually unnecessary because moves enable them; optional start code.
M18 / M84	M84	Disable steppers, optionally after a timeout or by axis.	Common final command after parking. Position becomes untrusted.
M75	M75	Start the print-job timer.	Optional start protocol.
M77	M77	Stop the print-job timer.	Optional end protocol.
M73	M73 P50	Set print progress when enabled.	Optional periodic generator output.
M117	M117 Printing	Display a message on the printer LCD.	Optional start/end status.
M118	M118 message	Send text to the host/serial interface.	Optional host diagnostics.

4. Diagnostic and safety commands

These commands are useful during commissioning, but most should not appear automatically in every generated file.

Command	Description	Policy
M114	Report the current tool position.	Manual test/host diagnostic.
M115	Report firmware name, version and capabilities.	Use when building/detecting a machine profile.
M119	Report endstop states.	Commissioning and homing diagnosis.
M503	Report active settings.	Capture limits/settings for a machine profile.
M500	Save current settings to EEPROM.	Do not emit automatically. It permanently changes printer configuration.
M501	Reload settings from EEPROM.	Maintenance/profile operation only.
M502	Load firmware defaults into RAM.	Do not emit automatically. May discard expected tuning until settings are restored.
M0 / M1	Stop and wait for user interaction.	Optional deliberate operator checkpoint.
M108	Break out of supported waiting loops.	Host emergency-parser operation, not normal file output.
M112	Emergency shutdown and halt.	Emergency host/operator action. Not a normal end command.
M410	Quick-stop planner motion.	Emergency/host action. Machine position may become invalid.
M999	Restart from a stopped state.	Manual recovery only after the cause is understood.

5. Common generated move patterns

Absolute XYZ with relative extrusion (G90 + M83)

```

G21                ; millimetres
G90                ; absolute XYZ
M83                ; relative E
G0 X20.000 Y20.000 Z0.300 F4200
G1 X60.000 Y20.000 Z0.300 E1.28450 F1800
G1 X60.000 Y60.000 Z0.300 E1.28450

```

Travel, retract, Z-hop, approach and recover

```
G1 E-1.00000 F1800      ; relative retract under M83
G0 Z1.300 F1200        ; raise from printing Z=0.300
G0 X100.000 Y80.000 F4200 ; travel without E
G0 Z0.300 F1200        ; return to printing height
G1 E1.00000 F1800      ; recover
```

Absolute extrusion (M82)

```
M82
G92 E0
G1 X20.000 Y20.000 Z0.300 E0.50000 F1800
G1 X40.000 Y20.000 Z0.300 E1.14225
G1 X60.000 Y20.000 Z0.300 E1.78450
```

When using `M82`, each value is the cumulative E coordinate. Never output a per-segment E amount directly.

Arc deposition when supported

```
G17      ; XY plane, if explicit plane selection is enabled/needed
G2 X80.000 Y60.000 I20.000 J0.000 E2.10000 F1800
```

Arc emission requires:

- verified `ARC_SUPPORT` in Marlin;
- planar arc geometry;
- correct clockwise/counter-clockwise direction;
- correct centre offsets or radius mode;
- full-circle and tolerance handling;
- linear `G1` fallback when any requirement is not satisfied.

6. FDM start protocol template

The values below are examples only. Replace temperatures, levelling code, purge path, feeds and coordinates with tested values for the exact printer.

```
; ===== CASA CRETA START - EXAMPLE FDM PROFILE =====
M117 Preparing
G21                ; millimetres
G90                ; absolute XYZ
M83                ; relative extrusion
M107               ; print fan off
M220 S100          ; feed override 100% - optional policy
M221 S100          ; flow override 100% - optional policy

M140 S60           ; begin bed heating, no wait
M104 S200          ; begin nozzle heating, no wait
G28                ; home approved axes
M420 S1            ; restore stored levelling if supported
M190 S60           ; wait for bed heating
M109 S200          ; wait for nozzle heating

G92 E0
G0 Z5.000 F1200
G0 X10.000 Y10.000 F4200
; Add a printer-approved purge/prime sequence here.
M75                ; optional print timer
M117 Printing
; ===== GENERATED MOTION FOLLOWS =====
```

Alternative levelling workflows may use `G29` instead of `M420 S1`. Use only the printer's known working sequence.

7. Clay/paste start protocol template

This assumes the material system maps deposition to Marlin's E axis. Pump, relay, pressure or custom-controller systems may need different commands.

```
; ===== CASA CRETA START - EXAMPLE CLAY/PASTE PROFILE =====
M117 Preparing clay job
G21                ; millimetres
G90                ; absolute XYZ
M83                ; relative process/extrusion value
M107               ; fan off unless process requires it
G28                ; machine-approved homing procedure
M302 S0            ; allow cold E motion - PROFILE MUST APPROVE
G92 E0
G0 Z10.000 F1200
; Add machine-approved pump/nozzle preparation here.
M75
M117 Printing
; ===== GENERATED MOTION FOLLOWS =====
```

Never place `M302 S0` in a generic FDM profile. Cold extrusion can strip filament, damage an extruder, or create unsafe unexpected motion. It belongs only in an explicitly approved non-thermal extrusion profile.

8. End protocol templates

FDM with relative E (M83)

```
; ===== CASA CRETA END - EXAMPLE FDM PROFILE =====
M400                ; complete queued motion
G1 E-2.00000 F1800  ; retract
G91
G0 Z10.000 F1200    ; relative Z lift
G90
G0 X10.000 Y200.000 F4200 ; approved park coordinate
M107                ; fan off
M104 S0             ; hotend off
M140 S0             ; bed off
M77                 ; optional stop timer
M117 Complete
M84                 ; disable steppers after parking
```

Clay/paste end

```
; ===== CASA CRETA END - EXAMPLE CLAY/PASTE PROFILE =====
M400                ; finish all deposition moves
G1 E-1.00000 F600   ; profile-defined pressure relief
G91
G0 Z20.000 F1200    ; safe relative lift
G90
G0 X10.000 Y200.000 F3000 ; approved park coordinate
M77
M117 Complete
M84
```

Do not copy example park coordinates to a real machine without confirming the build envelope, origin, kinematics and already-printed geometry.

9. Required generator validation

Before serialization, reject or warn on:

- invalid, null or zero-length curves;
- NaN, infinity or values outside numeric precision policy;
- incorrect Rhino-to-millimetre scale conversion;
- XYZ coordinates outside the machine envelope;
- Z below the safe minimum or above the machine maximum;
- deposition before homing/position establishment in the chosen protocol;
- extrusion during a travel transition;
- negative absolute E values under M82, unless deliberately supported;
- mismatched M82/M83 calculation mode;
- feedrates above axis/process limits;
- extrusion/flow above the process limit;
- temperatures outside the machine/material profile;
- unsupported G2/G3, G10/G11, G29, M420, tool or feature commands;
- a missing start or end protocol;
- unsafe first move to the first curve;
- unapproved cold extrusion;
- shutdown before M400 when queued motion must finish.

10. Command emission policy

Always emit or establish

G21
G90 or deliberately selected XYZ mode
M82 or M83
Known machine start protocol
Generated G0/G1 motion
Known machine end protocol

Emit only when enabled by MachineProfile

G2/G3 arcs
G10/G11 firmware retract
G29 / M420 bed levelling
M200 volumetric extrusion
M204/M205 motion overrides
M302 cold extrusion
Tn tool selection/change
G53 or workspace coordinate systems
M73/M75/M77 job reporting

Never emit automatically from the curve compiler

M500 EEPROM save
M502 factory defaults
M851 probe calibration changes
M92 steps-per-unit changes
M201/M203 machine-limit changes unless explicitly approved
M112 emergency stop as a normal end command
M999 restart
Firmware update, pin-control or board-debug commands

11. Minimal C# postprocessor state

The serializer should track at least:

UnitsMode	Millimetres / Inches
XyzPositionMode	Absolute / Relative
ExtrusionMode	Absolute / Relative
CurrentX, CurrentY, CurrentZ	
CurrentE	
CurrentFeedrate	
ActiveTool	
FanState	
HotendTarget	
BedTarget	
LevellingState	Unknown / Enabled / Disabled
SteppersTrusted	Unknown / Homed / Untrusted

Recommended internal pipeline:

```
RhinoCommon Curve
-> ToolpathPath
-> PlannedMove
-> validated MotionProgram
-> MarlinPostProcessor
-> GCodeBlock list
-> final text
```

Do not build G-code by concatenating strings directly inside the curve-sampling loop. First create typed moves, validate them, and only then serialize them.

12. Official references

- [Marlin G-code index](#)
- [G0/G1 linear moves](#)
- [G2/G3 arc moves](#)
- [G28 homing](#)
- [G90 absolute positioning](#)
- [G91 relative positioning](#)
- [G92 set position](#)
- [M82 absolute extrusion](#)
- [M83 relative extrusion](#)
- [M104 hotend temperature](#)
- [M109 wait for hotend](#)
- [M140 bed temperature](#)
- [M190 wait for bed](#)
- [M106 fan speed](#)
- [M204 acceleration](#)
- [M205 advanced motion settings](#)
- [M220 feedrate percentage](#)
- [M221 flow percentage](#)
- [M302 cold extrusion](#)
- [M400 finish moves](#)

Casa Creta - G_code_engine

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Revision: 2026-08-14